

Name: _____

Student ID: _____

If (the last 3 digits of your student ID) mod 5 = n , then you need to solve the n -th problem from the following:

a) Design Turing machines to perform the following operations (the leftmost bit is LSB) using the simulator provided (for manual and more details, see <https://schaetzc.github.io/tursi/>) (10)

0. Increment (add 1). The tape initially contains wc , where w is a binary number. Your TM should add 1 to w and write the sum to the right of the c .
1. Decrement (subtract 1). The tape initially contains wc , where w is a binary number. Your TM should subtract 1 from w and write the result to the right of the c .
2. Logical XOR. The tape initially contains w_1cw_2c , where w_1 and w_2 are binary numbers. Your TM should compute logical XOR of w_1 and w_2 and write the result to the right of the second c .
3. Logical AND. The tape initially contains w_1cw_2c , where w_1 and w_2 are binary numbers. Your TM should compute logical AND of w_1 and w_2 and write the result to the right of the second c .
4. Logical OR. The tape initially contains w_1cw_2c , where w_1 and w_2 are binary numbers. Your TM should compute logical OR of w_1 and w_2 and write the result to the right of the second c .

b) Describe an undecidable problem that has *not* been mentioned in class. Outline the history of how it was proved to be undecidable within two *handwritten* pages (you don't need to provide full proofs). (10)

Submission instructions:

Submit the *.tm file for the first problem and the scanned pages for the second problem in a zipped folder, named using your student ID, through moodle.

Deadline: 02/10/2026 (Friday) 11:59pm